

Supplementary Material for MoReAD: Learning Multi-View Semantic Representations for Unsupervised Anomaly Detection

Anonymous submission

A Additional Theoretical Analysis

A.1 Cross-Representation Responsibility and Routing

Entropy-regularized responsibility. For token i , let

$$e_{i,k} = \lambda_D d_{i,k}^D + \lambda_E d_{i,k}^E, \quad q_i \in \Delta^{K-1}. \quad (\text{A.1})$$

The responsibility follows from

$$q_i^* = \arg \min_{q_i \in \Delta^{K-1}} \sum_{k=1}^K q_{i,k} e_{i,k} + \tau_q \sum_{k=1}^K q_{i,k} \log q_{i,k}. \quad (\text{A.2})$$

Introducing η_i for $\sum_k q_{i,k} = 1$ gives

$$\mathcal{J}(q_i, \eta_i) = \sum_k q_{i,k} e_{i,k} + \tau_q \sum_k q_{i,k} \log q_{i,k} + \eta_i \left(\sum_k q_{i,k} - 1 \right), \quad (\text{A.3})$$

$$\frac{\partial \mathcal{J}}{\partial q_{i,k}} = e_{i,k} + \tau_q (1 + \log q_{i,k}) + \eta_i = 0, \quad (\text{A.4})$$

$$\log q_{i,k} = -\frac{e_{i,k}}{\tau_q} - \frac{\eta_i + \tau_q}{\tau_q}, \quad (\text{A.5})$$

$$q_{i,k} = \exp\left(-\frac{e_{i,k}}{\tau_q}\right) \exp\left(-\frac{\eta_i + \tau_q}{\tau_q}\right). \quad (\text{A.6})$$

Normalization cancels the second factor:

$$q_{i,k}^* = \frac{\exp(-e_{i,k}/\tau_q)}{\sum_{j=1}^K \exp(-e_{i,j}/\tau_q)}. \quad (\text{A.7})$$

If the minimum cost is unique,

$$\lim_{\tau_q \rightarrow 0^+} q_{i,k}^* = \mathbb{I}\left[k = \arg \min_j e_{i,j}\right], \quad (\text{A.8})$$

$$\lim_{\tau_q \rightarrow \infty} q_{i,k}^* = \frac{1}{K}. \quad (\text{A.9})$$

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Training-only perceptual supervision. Fixing the learned parameters θ , the inference variables satisfy

$$X \longrightarrow H \longrightarrow Y = T_\theta(H), \quad I(X; Y | H, \theta) = 0. \quad (\text{A.10})$$

Applying the chain rule in two orders,

$$I(X; H, Y | \theta) = I(X; H | \theta) + I(X; Y | H, \theta) \quad (\text{A.11})$$

$$= I(X; H | \theta), \quad (\text{A.12})$$

$$I(X; H, Y | \theta) = I(X; Y | \theta) + I(X; H | Y, \theta) \quad (\text{A.13})$$

$$\geq I(X; Y | \theta). \quad (\text{A.14})$$

Hence

$$I(X; Y | \theta) \leq I(X; H | \theta). \quad (\text{A.15})$$

EUPE can shape θ during training, but supplies no sample-specific variable to T_θ at inference.

Optimal DINO-only router. Let Q be the training responsibility and $p(H)$ the router. Its population objective is

$$p^* = \arg \min_p \mathbb{E}_{H, Q} D_{\text{KL}}(Q \| p(H)). \quad (\text{A.16})$$

For fixed $H = h$, define

$$a_k(h) = \mathbb{E}[Q_k | H = h], \quad \sum_k a_k(h) = 1. \quad (\text{A.17})$$

Then, with

$$\mathcal{L}_h(p) = \mathbb{E}[D_{\text{KL}}(Q \| p(h)) | H = h], \quad (\text{A.18})$$

$$\begin{aligned} \mathcal{L}_h(p) &= \mathbb{E}\left[\sum_k Q_k \log Q_k | h\right] \\ &\quad - \sum_k a_k(h) \log p_k(h) \end{aligned} \quad (\text{A.19})$$

$$= C(h) - \sum_k a_k(h) \log p_k(h). \quad (\text{A.20})$$

With multiplier ξ ,

$$\mathcal{R}(p, \xi) = - \sum_k a_k \log p_k + \xi \left(\sum_k p_k - 1 \right), \quad (\text{A.21})$$

$$\frac{\partial \mathcal{R}}{\partial p_k} = -\frac{a_k}{p_k} + \xi = 0, \quad (\text{A.22})$$

$$p_k = \frac{a_k}{\xi}, \quad (\text{A.23})$$

$$1 = \sum_k p_k = \frac{1}{\xi} \sum_k a_k = \frac{1}{\xi}. \quad (\text{A.24})$$

Therefore

$$p_k^*(h) = a_k(h) = \mathbb{E}[Q_k \mid H = h]. \quad (\text{A.25})$$

Moreover, Pinsker's inequality gives

$$\|q_i - p_i\|_1 \leq \sqrt{2D_{\text{KL}}(q_i \parallel p_i)}. \quad (\text{A.26})$$

If $\|\Delta_{i,k}\|_2 \leq M$, the correction error obeys

$$E_i = \gamma \sum_k (q_{i,k} - p_{i,k}) \Delta_{i,k}. \quad (\text{A.27})$$

$$\|E_i\|_2 \leq \gamma \sum_k |q_{i,k} - p_{i,k}| \|\Delta_{i,k}\|_2 \quad (\text{A.28})$$

$$\leq \gamma M \|q_i - p_i\|_1 \quad (\text{A.29})$$

$$\leq \gamma M \sqrt{2D_{\text{KL}}(q_i \parallel p_i)}. \quad (\text{A.30})$$

Top- n truncation. Let S_i contain the n largest entries of p_i and

$$\begin{aligned} m_i &= \sum_{k \in S_i} p_{i,k}, \\ \tilde{p}_{i,k} &= \begin{cases} p_{i,k}/m_i, & k \in S_i, \\ 0, & k \notin S_i. \end{cases} \end{aligned} \quad (\text{A.31})$$

Set

$$\begin{aligned} A_i &= \sum_{k \in S_i} p_{i,k} \Delta_{i,k}, \\ R_i &= \sum_{k \notin S_i} p_{i,k} \Delta_{i,k}. \end{aligned} \quad (\text{A.32})$$

The dense and sparse corrections are

$$\begin{aligned} \mu_i &= A_i + R_i, \\ \tilde{\mu}_i &= \frac{A_i}{m_i}. \end{aligned} \quad (\text{A.33})$$

Thus

$$\mu_i - \tilde{\mu}_i = R_i - \frac{1 - m_i}{m_i} A_i, \quad (\text{A.34})$$

$$\|R_i\|_2 \leq \sum_{k \notin S_i} p_{i,k} M = (1 - m_i)M, \quad (\text{A.35})$$

$$\|A_i\|_2 \leq \sum_{k \in S_i} p_{i,k} M = m_i M, \quad (\text{A.36})$$

$$\begin{aligned} \|\mu_i - \tilde{\mu}_i\|_2 &\leq (1 - m_i)M \\ &\quad + \frac{1 - m_i}{m_i} (m_i M) \end{aligned} \quad (\text{A.37})$$

$$= 2M(1 - m_i). \quad (\text{A.38})$$

After residual scaling,

$$\|\gamma \mu_i - \gamma \tilde{\mu}_i\|_2 \leq 2\gamma M(1 - m_i). \quad (\text{A.39})$$

A.2 Geometry-Preserving Residual Correction

Let $\|h\|_2 = 1$, $\bar{h} = (h + \delta)/\|h + \delta\|_2$, and $\|\delta\|_2 \leq \varepsilon < 1$. Writing $r = \|h + \delta\|_2$,

$$\begin{aligned} |r - 1| &= \left| \|h + \delta\|_2 - \|h\|_2 \right| \\ &\leq \|\delta\|_2 \leq \varepsilon, \end{aligned} \quad (\text{A.40})$$

$$r \geq 1 - \varepsilon. \quad (\text{A.41})$$

The normalized displacement satisfies

$$\|\bar{h} - h\|_2 = \left\| \frac{h + \delta}{r} - h \right\|_2 \quad (\text{A.42})$$

$$= \frac{\|\delta + (1 - r)h\|_2}{r} \quad (\text{A.43})$$

$$\leq \frac{\|\delta\|_2 + |1 - r|\|h\|_2}{r} \quad (\text{A.44})$$

$$\leq \frac{2\varepsilon}{1 - \varepsilon}. \quad (\text{A.45})$$

For two corrected tokens,

$$D_{ij} = \langle \bar{h}_i, \bar{h}_j \rangle - \langle h_i, h_j \rangle. \quad (\text{A.46})$$

$$\begin{aligned} D_{ij} &= \langle \bar{h}_i - h_i, \bar{h}_j \rangle \\ &\quad + \langle h_i, \bar{h}_j - h_j \rangle, \end{aligned} \quad (\text{A.47})$$

$$\begin{aligned} |D_{ij}| &\leq \|\bar{h}_i - h_i\|_2 \|\bar{h}_j\|_2 \\ &\quad + \|h_i\|_2 \|\bar{h}_j - h_j\|_2 \end{aligned} \quad (\text{A.48})$$

$$\leq \frac{4\varepsilon}{1 - \varepsilon}. \quad (\text{A.49})$$

Consequently, under the same uniform residual bound,

$$\begin{aligned} \text{Geo} &= \mathbb{E}_{i,j} [\cos(\bar{h}_i, \bar{h}_j) - \cos(h_i, h_j)] \\ &\leq \frac{4\varepsilon}{1 - \varepsilon}. \end{aligned} \quad (\text{A.50})$$

A.3 Conditional Recovery and CBTM Capacity

Ambiguity of deterministic recovery. Let $Y = X_{i-1}$, $X = X_i$, and

$$\mu(x) = \mathbb{E}[Y \mid X = x]. \quad (\text{A.51})$$

For a deterministic recovery $u(x)$,

$$\mathcal{R}_x(u) = \mathbb{E}[\|Y - u\|_2^2 \mid X = x]. \quad (\text{A.52})$$

$$\mathcal{R}_x(u) = \mathbb{E}[\|Y - \mu + \mu - u\|_2^2 \mid x] \quad (\text{A.53})$$

$$\begin{aligned} &= \mathbb{E}[\|Y - \mu\|_2^2 \mid x] \\ &\quad + 2(\mu - u)^\top \mathbb{E}[Y - \mu \mid x] \\ &\quad + \|\mu - u\|_2^2 \end{aligned} \quad (\text{A.54})$$

$$= \mathbb{E}[\|Y - \mu\|_2^2 \mid x] + \|\mu - u\|_2^2. \quad (\text{A.55})$$

Therefore

$$u^*(x) = \mu(x) = \mathbb{E}[Y \mid X = x]. \quad (\text{A.56})$$

For a conditional mixture with weights $\pi_m(x)$ and component means $\mu_m(x)$,

$$u^*(x) = \sum_{m=1}^M \pi_m(x) \mu_m(x), \quad (\text{A.57})$$

which may lie between valid recovery modes. Cosine reconstruction gives the corresponding angular form of this ambiguity.

Rank of the coefficient channel. For token t , write

$$\begin{aligned} B_t(\hat{x}_i) &\in \mathbb{R}^{K_b \times C}, \\ \alpha_t(x_{i-1,t}) &\in \mathbb{R}^{K_b}. \end{aligned} \quad (\text{A.58})$$

and

$$\begin{aligned} \Delta_t &= B_t(\hat{x}_i)^\top \alpha_t(x_{i-1,t}), \\ \Delta_t &\in \mathbb{R}^C. \end{aligned} \quad (\text{A.59})$$

Holding the decoder state and hence B_t fixed,

$$r_B = \text{rank}(B_t^\top), \quad r_\alpha = \text{rank}(J_{\alpha,t}). \quad (\text{A.60})$$

$$J_{\text{CBTM},t} = \frac{\partial \Delta_t}{\partial x_{i-1,t}} \quad (\text{A.61})$$

$$= B_t^\top \frac{\partial \alpha_t}{\partial x_{i-1,t}} \quad (\text{A.62})$$

$$= B_t^\top J_{\alpha,t}, \quad (\text{A.63})$$

$$\text{rank}(J_{\text{CBTM},t}) \leq \min\{r_B, r_\alpha\} \quad (\text{A.64})$$

$$\leq K_b. \quad (\text{A.65})$$

In contrast, a direct skip has

$$\begin{aligned} \Delta_t^{\text{skip}} &= x_{i-1,t}, \\ J_{\text{skip},t} &= I_C, \\ \text{rank}(J_{\text{skip},t}) &= C. \end{aligned} \quad (\text{A.66})$$

Thus, for $K_b < C$,

$$\begin{aligned} \frac{\text{rank}(J_{\text{CBTM},t})}{\text{rank}(J_{\text{skip},t})} &\leq \frac{K_b}{C}, \\ NK_b &< NC. \end{aligned} \quad (\text{A.67})$$

This establishes a per-token bandwidth bound, not zero anomaly information.

For perturbation a_t , set $x = x_{i-1,t}$ and assume $\|B_t\|_2 \leq \beta$ and an L_α -Lipschitz predictor. With B_t fixed,

$$\delta \Delta_t = \Delta_t(x + a_t) - \Delta_t(x). \quad (\text{A.68})$$

$$\delta \alpha_t = \alpha_t(x + a_t) - \alpha_t(x). \quad (\text{A.69})$$

$$\delta \Delta_t = B_t^\top \delta \alpha_t, \quad (\text{A.70})$$

$$\|\delta \Delta_t\|_2 \leq \|B_t\|_2 \|\delta \alpha_t\|_2 \quad (\text{A.71})$$

$$\leq \beta L_\alpha \|a_t\|_2. \quad (\text{A.72})$$

A direct skip instead gives

$$\begin{aligned} \delta \Delta_t^{\text{skip}} &= \Delta_t^{\text{skip}}(x + a_t) - \Delta_t^{\text{skip}}(x), \\ \|\delta \Delta_t^{\text{skip}}\|_2 &= \|a_t\|_2. \end{aligned} \quad (\text{A.73})$$

Gated transition magnitude. Let

$$\begin{aligned} g_{t,k}^{\text{soft}} &= \sigma(\ell_{t,k}), \\ g_{t,k}^{\text{hard}} &= \mathbb{I}[g_{t,k}^{\text{soft}} \geq 0.5]. \end{aligned} \quad (\text{A.74})$$

Training uses the straight-through form

$$\begin{aligned} \tilde{g}_{t,k} &= g_{t,k}^{\text{soft}} \\ &+ \text{sg}(g_{t,k}^{\text{hard}} - g_{t,k}^{\text{soft}}). \end{aligned} \quad (\text{A.75})$$

while inference uses $g_{t,k}^{\text{hard}}$. Hence

$$\begin{aligned} \alpha_{t,k} &= g_{t,k} v_{t,k}, \\ |v_{t,k}| &\leq a, \quad \|b_{t,k}\|_2 \leq b. \end{aligned} \quad (\text{A.76})$$

If $s_t = |\{k : g_{t,k} > 0\}|$, then

$$\|\Delta_t\|_2 = \left\| \sum_{k=1}^{K_b} \alpha_{t,k} b_{t,k} \right\|_2 \quad (\text{A.77})$$

$$\leq \sum_{k=1}^{K_b} |\alpha_{t,k}| \|b_{t,k}\|_2 \quad (\text{A.78})$$

$$\leq s_t a b \leq K_b a b. \quad (\text{A.79})$$

For orthonormal basis rows,

$$\|\Delta_t\|_2^2 = \alpha_t^\top B_t B_t^\top \alpha_t \quad (\text{A.80})$$

$$= \alpha_t^\top I_{K_b} \alpha_t = \|\alpha_t\|_2^2. \quad (\text{A.81})$$

Orthogonality regularization. Let the rows of $\tilde{B}_t$ be unit-normalized and $G_t = \tilde{B}_t \tilde{B}_t^\top$. Since $(G_t)_{kk} = 1$,

$$\begin{aligned} \|G_t - I_{K_b}\|_F^2 &= \sum_k ((G_t)_{kk} - 1)^2 \\ &+ \sum_{k \neq j} (G_t)_{kj}^2 \end{aligned} \quad (\text{A.82})$$

$$= \sum_{k \neq j} \langle \tilde{b}_{t,k}, \tilde{b}_{t,j} \rangle^2. \quad (\text{A.83})$$

At zero loss,

$$\begin{aligned} G_t = I_{K_b} &\implies \sigma_j(\tilde{B}_t) = 1, \quad j = 1, \dots, K_b \\ &\implies \text{rank}(\tilde{B}_t) = K_b. \end{aligned} \quad (\text{A.84})$$

With

$$\begin{aligned} \rho_j &= \frac{\sigma_j}{\sum_l \sigma_l}, \\ \text{erank}(\tilde{B}_t) &= \exp \left(- \sum_j \rho_j \log \rho_j \right). \end{aligned} \quad (\text{A.85})$$

equal singular values attain

$$\text{erank}(\tilde{B}_t) = K_b. \quad (\text{A.86})$$

A.4 Capacity and Diagnostic Quantities

Parameter and operation counts. For an expert $\mathbb{R}^C \rightarrow \mathbb{R}^r \rightarrow \mathbb{R}^C$, ignoring biases,

$$P_{\text{expert}} = Cr + rC = 2Cr, \quad (\text{A.87})$$

$$P_{K_e \text{ experts}} = 2K_e Cr. \quad (\text{A.88})$$

The multiply-accumulate counts per token are

$$\text{MAC}_{\text{dense}} = 2K_e Cr, \quad (\text{A.89})$$

$$\text{MAC}_{\text{Top-}n} = 2nCr, \quad (\text{A.90})$$

$$\frac{\text{MAC}_{\text{Top-}n}}{\text{MAC}_{\text{dense}}} = \frac{n}{K_e}. \quad (\text{A.91})$$

Algorithm 1 Frozen DINO/EUPE feature extraction

Require: image x ; frozen DINO encoder E_D ; frozen EUPE encoder E_E

Require: DINO shallow layers $\mathcal{S}_D$, DINO deep layers $\mathcal{D}_D$, EUPE semantic layers $\mathcal{D}_E$

- 1: $z^D \leftarrow \text{Mean}_{\ell \in \mathcal{S}_D} E_D^\ell(x)$
- 2: $z^{D,+} \leftarrow \text{Mean}_{\ell \in \mathcal{D}_D} E_D^\ell(x)$
- 3: **if** training or privileged diagnostic evaluation **then**
- 4: $z^{E,+} \leftarrow \text{ResizeAlign}(\text{Mean}_{\ell \in \mathcal{D}_E} E_E^\ell(x), \text{grid}(z^{D,+}))$
- 5: **else**
- 6: $z^{E,+} \leftarrow \emptyset$
- 7: **end if**
- 8: remove DINO register tokens and keep the CLS token identity path
- 9: $z^{\text{seed}} \leftarrow \frac{1}{2}(z^D + z^{D,+})$
- 10: **return** $z^{\text{seed}}, z^D, z^{D,+}, z^{E,+}$

For CBTM aggregation,

$$\begin{aligned} \text{MAC}_{\text{CBTM}} &= NK_b C, \\ \frac{NK_b}{NC} &= \frac{K_b}{C}. \end{aligned} \quad (\text{A.92})$$

Hence low parameter count does not imply low computation when the learned operators are evaluated densely over N tokens.

Anomaly-preservation gap. Let d be the reconstruction distance and

$$\text{APG} = \mathbb{E}[d \mid A] - \mathbb{E}[d \mid N]. \quad (\text{A.93})$$

For $d' = d + \delta_d$,

$$\text{APG}' = \mathbb{E}[d + \delta_d \mid A] - \mathbb{E}[d + \delta_d \mid N] \quad (\text{A.94})$$

$$= \text{APG} + \mathbb{E}[\delta_d \mid A] - \mathbb{E}[\delta_d \mid N], \quad (\text{A.95})$$

so

$$\begin{aligned} \text{APG}' > \text{APG} &\iff \mathbb{E}[\delta_d \mid A] \\ &> \mathbb{E}[\delta_d \mid N]. \end{aligned} \quad (\text{A.96})$$

In particular, reducing normal residuals more strongly than anomaly residuals increases APG.

B Training and Inference Pseudocode

Algorithms 1–5 summarize the implemented MoReAD pipeline. $\text{sg}(\cdot)$ denotes stop-gradient. $\text{NormTop}_n(\cdot)$ keeps the largest n entries along the expert dimension, sets the rest to zero, and renormalizes the remaining weights. In the default configuration, $K_e = 4$, $n = 2$, $K_b = 8$, $\tau_p = 0.5$, and $\tau_q = 0.003$.

C Reproduction Details

Unless stated otherwise, all core experiments use the same multiclass normal-only training protocol. DINO and EUPE are frozen. The bottleneck, decoder, CR-MoE router/adapters/proxy heads, and CBTM modules are optimized jointly. EUPE is used only as privileged supervision during training and diagnostic privileged evaluation; the deployed model uses DINO only.

Algorithm 2 CR-MoE semantic bottleneck

Require: bottleneck tokens h ; patch start c ; patch grid (H, W)

Require: low-rank adapters $\{A_k\}_{k=1}^{K_e}$; router G ; proxies P_D, P_E

Require: teacher targets $z^{D,+}$ and optional $z^{E,+}$

- 1: reshape patch tokens h_c into $h^{\text{map}} \in \mathbb{R}^{B \times C \times H \times W}$
- 2: $p \leftarrow \text{softmax}(G(h^{\text{map}})/\tau_p)$
- 3: **if** training and dense warm-up is active **then**
- 4: $g \leftarrow p$
- 5: **else**
- 6: $g \leftarrow \text{NormTop}_n(p)$
- 7: **end if**
- 8: **for** $k = 1, \dots, K_e$ **do**
- 9: $\Delta_k \leftarrow A_k(h^{\text{map}})$
- 10: **end for**
- 11: $\gamma \leftarrow \gamma_{\max} \sigma(a_\gamma)$
- 12: $\tilde{h}^{\text{map}} \leftarrow h^{\text{map}} + \gamma \sum_{k=1}^{K_e} g_k \Delta_k$
- 13: $\tilde{h} \leftarrow \text{flatten } \tilde{h}^{\text{map}}$ and concatenate unchanged non-patch tokens
- 14: **if** no teacher target is available **then**
- 15: **return** $\tilde{h}$ and routing record (p, g, γ)
- 16: **end if**
- 17: **for** $k = 1, \dots, K_e$ **do**
- 18: $h_k^{\text{map}} \leftarrow h^{\text{map}} + \gamma \Delta_k$
- 19: $d_k^D \leftarrow 1 - \cos(P_D(h_k^{\text{map}}), z^{D,+})$
- 20: **if** $z^{E,+}$ exists **then**
- 21: $d_k^E \leftarrow 1 - \cos(P_E(h_k^{\text{map}}), z^{E,+})$
- 22: **else**
- 23: $d_k^E \leftarrow 0$
- 24: **end if**
- 25: $e_k \leftarrow \lambda_D d_k^D + \lambda_E d_k^E$
- 26: **end for**
- 27: $q \leftarrow \text{softmax}(-e/\tau_q)$
- 28: **if** privileged evaluation **then**
- 29: $g \leftarrow \text{NormTop}_n(q)$
- 30: recompute $\tilde{h}$ using g
- 31: **end if**
- 32: **return** $\tilde{h}$ and routing record (p, g, q, e, γ)

C.1 Data and Feature Extraction

C.2 Architecture

C.3 Loss and Optimization

C.4 Inference and Runtime

MVTec AD2. MVTec AD2 experiments use a separate sliding-window evaluation and submission pipeline because the images are much larger than the core 392×392 setting. Unless explicitly stated for an AD2 experiment, the model modules and losses above are unchanged, while the image loading, tiling, stitching, and optional public-set post-processing follow the AD2-specific configuration saved with each run.

Algorithm 3 One CBTM transition stage

Require: decoder prediction $\hat{z}_s$; DINO target (encoder feature) z_s
Require: conditional basis generator G_s^B ; coefficient predictor G_s^C

- 1: $B_s \leftarrow G_s^B(\hat{z}_s)$ { B_s reads only the current decoder prediction}
- 2: $B_s \leftarrow \text{Normalize}(B_s)$
- 3: $(\ell_s, v_s) \leftarrow G_s^C(z_s)$ {coefficients read the corresponding encoder feature z_s }
- 4: $\rho_s \leftarrow \sigma(\ell_s)$
- 5: **if** training **then**
- 6: $m_s \leftarrow \text{BernoulliST}(\rho_s)$
- 7: **else**
- 8: $m_s \leftarrow \mathbb{I}[\rho_s \geq 0.5]$
- 9: **end if**
- 10: $\alpha_s \leftarrow m_s \odot \alpha_{\max} \tanh(v_s)$
- 11: $\Delta_s^{\text{CBTM}} \leftarrow \sum_{j=1}^{K_b} \alpha_{s,j} B_{s,j}$
- 12: $\hat{z}_s^{\text{CBTM}} \leftarrow \hat{z}_s + \Delta_s^{\text{CBTM}}$
- 13: $r_s \leftarrow z_s - \hat{z}_s$
- 14: $a_s \leftarrow \langle \text{sg}(r_s), \text{sg}(B_s) \rangle$ { a_s is used only for diagnostics/alignment, not as an input}
- 15: **return** $\hat{z}_s^{\text{CBTM}}$ and support record $(r_s, \Delta_s^{\text{CBTM}}, B_s, \alpha_s, \rho_s, m_s, a_s)$

D Additional Experiments

This section expands the aggregate comparison with category-level results, measured baseline checks, component transfer, mechanism controls, routing statistics, and numerical sensitivity values. All metrics follow Section C. Values are reported in percent unless the table explicitly states otherwise.

D.1 Category-Level Results

Table 11 gives the category-level MoReAD results used to form the dataset means in the main comparison table. The table is split into two blocks only for page layout; the macro mean is computed over categories within each dataset. Dataset means are computed from the unrounded per-category records and then rounded for display.

D.2 Measured Baseline Checks

Table 12 lists the DINO reconstruction baseline runs that were directly measured in the current codebase. They check that the implementation, scoring, and dataset adapters produce the expected baseline range.

D.3 Component Transfer

The component-transfer study separates CR-MoE and CBTM on the larger or less-standard benchmarks. Every row keeps the same backbone, decoder capacity, training schedule, and post-processing.

D.4 Coefficient-Channel Controls

Table 14 varies the information available to the coefficient predictor. The deployed design uses encoder-side features

Algorithm 4 MoReAD training on normal images

Require: normal image batch x ; frozen encoders E_D, E_E
Require: bottleneck B_θ ; decoder blocks $\{D_t\}_{t=1}^T$; CR-MoE $\mathcal{R}_\phi$; CBTM modules $\{\mathcal{C}_s\}_{s=1}^S$

- 1: $z^{\text{seed}}, z^{D,+}, z^{E,+} \leftarrow \text{FeatureExtract}(x)$
- 2: $h \leftarrow B_\theta(z^{\text{seed}})$
- 3: $\tilde{h}, \mathcal{O}_{\text{CR}} \leftarrow \mathcal{R}_\phi(h, z^{D,+}, z^{E,+})$
- 4: run the chained decoder from $\tilde{h}$ through all decoder blocks
- 5: reverse decoder block outputs and average each decoder group
- 6: obtain loose predictions $\{\hat{z}_s\}_{s=1}^S$ for DINO target groups $\{z_s\}_{s=1}^S$
- 7: **for** $s = 1, \dots, S$ **do**
- 8: $\hat{z}_s^{\text{CBTM}}, \mathcal{O}_s^{\text{CBTM}} \leftarrow \text{CBTMStage}(\hat{z}_s, z_s; \mathcal{C}_s)$
- 9: **end for**
- 10: $\mathcal{L}_{\text{rec}} \leftarrow \frac{1}{S} \sum_s [1 - \cos(\text{sg}(z_s), \hat{z}_s)]$
- 11: $\mathcal{L}_{\text{fit}} \leftarrow \frac{1}{S} \sum_s [1 - \cos(\text{sg}(z_s), \hat{z}_s^{\text{CBTM}})]$
- 12: $\mathcal{L}_{\text{act}} \leftarrow \frac{1}{S} \sum_s \text{Mean}(\rho_s)$
- 13: $\mathcal{L}_{\text{mag}} \leftarrow \frac{1}{S} \sum_s \text{Mean}(|\alpha_s|)$
- 14: $\mathcal{L}_{\text{orth}} \leftarrow \frac{1}{S} \sum_s \|B_s B_s^\top - I\|_F^2$
- 15: $\mathcal{L}_{\text{mix}} \leftarrow \frac{1}{S} \sum_s \text{Mean}(|\Delta_s^{\text{CBTM}}|)$
- 16: $\mathcal{L}_{\text{CBTM}} \leftarrow \lambda_{\text{fit}} \mathcal{L}_{\text{fit}} + \lambda_{\text{act}} \mathcal{L}_{\text{act}} + \lambda_{\text{mag}} \mathcal{L}_{\text{mag}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}} + \lambda_{\text{mix}} \mathcal{L}_{\text{mix}}$
- 17: $q \leftarrow \text{sg}(\mathcal{O}_{\text{CR}}.q)$; $p \leftarrow \mathcal{O}_{\text{CR}}.p$; $e \leftarrow \mathcal{O}_{\text{CR}}.e$
- 18: $\mathcal{L}_{\text{branch}} \leftarrow \text{Mean}_{b,i} \sum_k q_{bik} e_{bik}$
- 19: $\mathcal{L}_{\text{router}} \leftarrow \text{Mean}_{b,i} D_{\text{KL}}(q_{bi} \| p_{bi})$
- 20: $\bar{p}_k \leftarrow \text{Mean}_{b,i} p_{bik}$
- 21: $\mathcal{L}_{\text{bal}} \leftarrow \frac{1}{K_e} \sum_k (\bar{p}_k - 1/K_e)^2$
- 22: $\mathcal{L}_{\text{CR}} \leftarrow \lambda_{\text{branch}} \mathcal{L}_{\text{branch}} + \lambda_{\text{router}} \mathcal{L}_{\text{router}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}}$
- 23: $\mathcal{L} \leftarrow \lambda_{\text{base}} \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{CBTM}} + \mathcal{L}_{\text{CR}}$
- 24: update trainable bottleneck, decoder, CR-MoE, and CBTM parameters using $\nabla \mathcal{L}$

only through a bounded, gated low-rank coefficient channel, rather than direct feature copying, preserving a constrained anomaly measurement space.

D.5 Routing Diagnostics

Table 15 measures whether the DINO router retains the training-time responsibility while maintaining distinct expert usage. Entropies and agreement are first averaged over tokens.

D.6 Numerical Sensitivity Results

Table 16 reports numerical sensitivity values for the VisA-Real-IAD mean, complementing the main sensitivity plot; within each group, one parameter varies while the remaining settings stay at $K_e = 4$, Top- $n = 2$, and $K_b = 8$.

D.7 Additional Efficiency Results

Table 17 extends the main efficiency comparison with training cost. Measurements follow the runtime protocol in Table 9.

Algorithm 5 MoReAD DINO-only inference

Require: test image x ; frozen DINO encoder E_D

Require: trained bottleneck, CR-MoE router/adapters, decoder, CBTM modules

```
1: extract  $z^D$  and  $z^{D,+}$  using DINO only
2:  $z^{\text{seed}} \leftarrow \frac{1}{2}(z^D + z^{D,+})$ 
3:  $h \leftarrow B_\theta(z^{\text{seed}})$ 
4:  $p \leftarrow \text{softmax}(G(h)/\tau_p)$ 
5:  $g \leftarrow \text{NormTop}_n(p)$ 
6:  $\tilde{h} \leftarrow h + \gamma \sum_k g_k A_k(h)$ 
7: run the chained decoder from  $\tilde{h}$  and form loose predictions  $\{\hat{z}_s\}_{s=1}^S$ 
8: for  $s = 1, \dots, S$  do
9:    $\hat{z}_s^{\text{CBTM}} \leftarrow \text{CBTMStage}(\hat{z}_s, z_s)$ 
10:   $A_s \leftarrow 1 - \cos(z_s, \hat{z}_s^{\text{CBTM}})$ 
11: end for
12:  $A_{\text{raw}} \leftarrow \frac{1}{S} \sum_s A_s$ 
13:  $A \leftarrow \text{Gaussian}_{k=5, \sigma=4}(\text{Resize}_{256}(A_{\text{raw}}))$ 
14:  $s_{\text{image}} \leftarrow \text{Mean}(\text{Top}_{1\%}(A))$ 
15: return anomaly map  $A$  and image score  $s_{\text{image}}$ 
```

E Additional Visualizations

This section complements the numerical results with localization maps and internal mechanism visualizations. Anomaly maps are displayed with percentile-based scaling for readability, while all quantitative comparisons use the raw maps.

E.1 Qualitative Localization

Figure 1 compares selected VisA cases where MoReAD improves over Dinomaly. To emphasize localization precision rather than response magnitude, the prediction rows show the top-response pixels with the same area as the ground-truth defect.

E.2 Expert Routing

Figure 2 visualizes routing diagnostics from the measured MVTEC AD and VisA checkpoints. Lower KL and higher Top- n Jaccard indicate that the DINO-only router retains the training-time responsibility after EUPE is removed.

E.3 Conditional Basis Transitions

Figure 3 links CBTM activation statistics to localization behavior on the measured MVTEC–VisA diagnostic subset. The learned gates concentrate the support correction on a small subset of deep-stage tokens; coefficient magnitude is therefore reported both over all tokens and over active tokens.

E.4 Representative Improvement Cases

Figure 4 collects representative high-gain cases from the measured VisA checkpoints. Samples are selected by the largest positive per-sample pixel-AP gain over Dinomaly and are used only for qualitative inspection.

Configuration	Setting
DINO backbone	DINOv2 ViT-B/14 Reg4, 768 channels
DINO identifier	dinov2reg_vit_base_14
EUPE backbone	BiliSakura/EUPE-ViT-B, 768 channels
EUPE checkpoint path	tmp/eupe_cache/BiliSakura_EUPE-ViT-B
Trainable encoders	none
Image transform	resize to 448×448 , convert to tensor, center crop to 392×392
Normalization	ImageNet mean (0.485, 0.456, 0.406) and std (0.229, 0.224, 0.225)
Mask transform	resize to 448×448 , center crop to 392×392 , tensor mask
DINO patch grid	28×28 after removing register tokens
DINO shallow group	arithmetic mean of layers {2, 3, 4, 5}
DINO deep group	arithmetic mean of layers {6, 7, 8, 9}
EUPE semantic group	arithmetic mean of layers {6, 7, 8, 9}, bilinearly aligned to DINO grid
Deployed seed	$\frac{1}{2}(z_{\text{shallow}}^D + z_{\text{deep}}^D)$
Loose targets	two DINO groups: shallow and deep

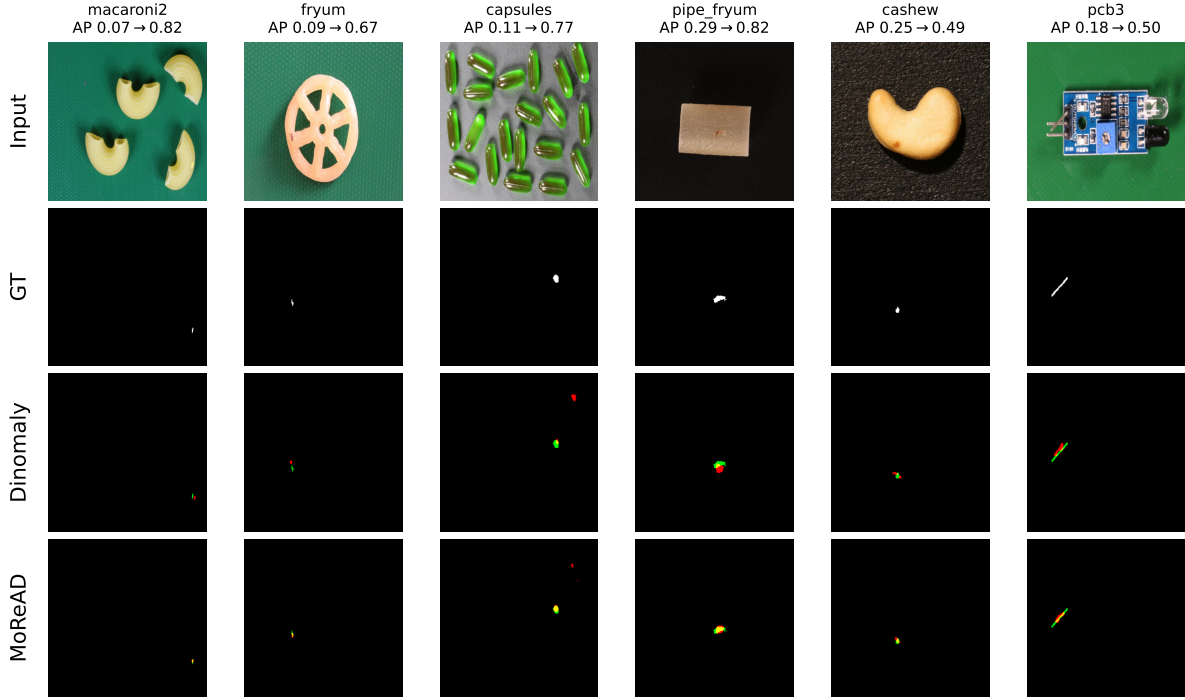
Table 1: Backbones, transforms, and feature extraction used in the core experiments.

Configuration	Setting
Training regime	one normal-only multiclass model per dataset
Core datasets	MVTec AD and VisA
MVTec AD classes	all 15 official categories
VisA classes	all 12 official categories in the local 1-class layout
Seeds	0 for the reported core run
Evaluation schedule	final epoch only for the core ablation
Checkpoint rule	final evaluated checkpoint; save_best_aupro is equivalent because only epoch 10 is evaluated
Primary reported metrics	I-F1, P-AUPRO@0.05, and P-AP from the offline unified evaluator
Runtime profiling input	batch size 1, $3 \times 392 \times 392$, AMP-FP16

Table 2: Dataset and evaluation protocol for the core reproduction setting.

Component	Setting
Bottleneck	MLP bottleneck, hidden width $4C = 3072$, dropout 0.2
Decoder	8 transformer blocks with linear attention
Loose blocks	2 target groups, 4 decoder outputs averaged per group
Decoder path	chained deep-to-shallow decoding
Prediction adapters	disabled
Deep auxiliary adapters	disabled in the final A8 run
Score type	support_weighted

Table 3: Shared reconstruction architecture. Here $C = 768$.



Real VisA outputs selected by top-area localization improvement. In prediction rows, red denotes the top-|GT| predicted pixels, green denotes GT, and yellow denotes overlap.

Figure 1: Additional qualitative improvement cases on VisA. Rows show the input, ground truth, Dinomaly, and MoReAD. In prediction rows, red denotes the top-|GT| predicted pixels, green denotes ground truth, and yellow denotes overlap. Titles report the per-sample pixel-AP of the DINO reconstruction model in Dinomaly and MoReAD, respectively.

Hyperparameter	Setting
Number of experts K_e	4
Active experts	Top-2
Expert rank	64
Expert mapping	1×1 conv 768 $\rightarrow$ 64, GELU, 1×1 conv 64 $\rightarrow$ 768
Expert output initialization	zero-initialized output projection
Router	GroupNorm(1, C) followed by 1×1 conv to K_e logits
Router initialization	zero weight and zero bias
Router temperature τ_p	0.5
Responsibility temperature τ_q	0.003
DINO cost weight λ_D	1.0
EUPE cost weight λ_E	0.15
Proxy heads	separate 1×1 DINO and EUPE proxy convolutions
Dense-routing warm-up	epochs 1–2 use dense router probabilities
Sparse routing	Top-2 starts from epoch 3 and is always used at deployment
Residual scale	$\gamma = 0.5\sigma(a_\gamma)$
Scale initialization / maximum	0.1 / 0.5
Deployment	DINO router and adapters only; EUPE and proxy heads removed

Table 4: CR-MoE configuration.

Hyperparameter	Setting
Support type	single conditional CBTM
Number of bases K_b	8 per transition stage
Basis adapter rank	8
Transition stages	2 loose decoder stages
Basis input	current decoder prediction $\hat{z}_s$
Basis generator	per-basis LayerNorm, Linear(768, 8), GELU, Linear(8, 768), then ℓ_2 normalization
Coefficient input	corresponding encoder feature z_s
Coefficient MLP	LayerNorm and two hidden GELU layers, hidden width $\max(0.5C, 128) = 384$
Coefficient output	$2K_b$ channels split into gate logits and amplitudes
Gate probability	sigmoid of coefficient logits
Training gate	Bernoulli straight-through sample from the gate probability
Inference gate	binary threshold at 0.5
Coefficient amplitude	$2 \tanh(\cdot)$
Residual correction	$\hat{z}_s^{\text{CBTM}} = \hat{z}_s + \sum_j \alpha_{s,j} B_{s,j}$
Target usage	encoder features select bounded coefficients, but are not directly copied into the reconstruction

Table 5: CBTM configuration.

Loss term	Value / definition
Base reconstruction λ_{base}	1.0
Base distance	mean patch-token cosine distance to stopped DINO group target
Hard mining	disabled by $p = 0$
CBTM fit λ_{fit}	1.0
CBTM normalized MSE λ_{mse}	0.0
CBTM activation λ_{act}	10^{-3} on mean gate probability
CBTM magnitude λ_{mag}	10^{-3} on mean absolute coefficient
CBTM orthogonality λ_{orth}	10^{-3} on $\ BB^T - I\ _F^2$
Orthogonality token sample	256 patch tokens per stage
Orthogonality type	Gram penalty
CBTM correction magnitude λ_{mix}	10^{-3} on mean absolute support correction
Expert minimum-usage penalty	10^{-4} , threshold 0.02; inactive for single-support A8
CR branch λ_{branch}	0.1
Router KL λ_{router}	0.1
Router balance λ_{bal}	10^{-3} toward uniform mean router usage

Table 6: Loss weights and reductions. All patch-token losses exclude the CLS token.

Training configuration	Setting
Optimizer	StableAdamW
Learning rate	2×10^{-3}
Final learning rate	2×10^{-4}
Scheduler	linear warm-up followed by cosine decay
Warm-up	100 optimizer steps
Weight decay	10^{-4}
Adam betas	(0.9, 0.999)
Adam epsilon	10^{-10}
AMSGrad	enabled
Gradient clipping	global norm 0.1
Training stage	joint training of transition, CR-MoE, and CBTM modules
Epochs	10
GPU(s)	single NVIDIA GeForce RTX 5090
Batch size	24
Eval batch size	4
Data workers	4 train workers and 4 eval workers
Distributed backend	single-GPU (DDP available)

Table 7: Optimization configuration.

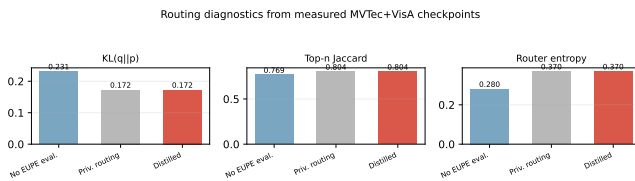


Figure 2: Routing diagnostics for CR-MoE. The distilled router reduces teacher-router KL while retaining high Top- n agreement.

Inference configuration	Setting
Inference encoders	DINO only
Routing	DINO-conditioned router with Top-2 normalized expert weights
Reconstruction map	cosine residual between DINO target group and CBTM-corrected prediction
Stage aggregation	equal average over the two loose stages
Metric map size	resize anomaly maps to 256×256
Ground-truth map size	nearest-neighbor resize to 256×256
Smoothing	Gaussian kernel 5×5 , $\sigma = 4$, applied after resizing
Image-level score	mean of the largest 1% anomaly-map values
Pixel metric	P-AP and P-AUPRO@0.05 from cached prediction arrays
Image metric	maximum-F1 image threshold search in the unified evaluator

Table 8: Anomaly scoring and post-processing.

Runtime/profiling configuration	Setting
Precision	AMP-FP16 for deployment profiling
Profile input	batch size 1, $3 \times 392 \times 392$
Warm-up / timed iterations	10 / 30
Deployment parameters	EUPE encoder and CR-MoE proxy heads excluded
Latency/FPS	CUDA-synchronized mean over timed iterations
Peak memory	maximum CUDA allocated memory during deployment forward
GFLOPs	measured on the deployed forward graph at 392×392

Table 9: Runtime measurement protocol.

Environment	Setting
Operating system	Ubuntu 22.04
Python	3.12
PyTorch	2.5.1
CUDA	12.4
GPU	single NVIDIA GeForce RTX 5090, 32 GB memory
CPU	25 Intel Xeon Platinum 8470Q vCPUs
Host memory	90 GB

Table 10: Software and hardware environment recorded for the core run.

Dataset	Category	I-F1	P-AUPRO@0.05	P-AP	Dataset	Category	I-F1	P-AUPRO@0.05	P-AP
MVTec AD	carpet	100.0	85.3	82.0	Real-IAD	button_battery	83.6	86.7	63.5
MVTec AD	grid	100.0	98.0	87.5	Real-IAD	end_cap	87.8	85.9	43.0
MVTec AD	leather	100.0	93.0	61.8	Real-IAD	eraser	77.2	87.0	58.1
MVTec AD	tile	99.5	71.0	60.3	Real-IAD	fire_hood	84.8	75.9	58.6
MVTec AD	wood	98.0	83.0	69.6	Real-IAD	mint	91.0	78.5	46.2
MVTec AD	bottle	98.3	98.0	82.6	Real-IAD	mounts	92.3	84.6	44.4
MVTec AD	cable	100.0	83.8	61.6	Real-IAD	pcb	86.3	78.8	66.5
MVTec AD	capsule	100.0	86.8	92.1	Real-IAD	phone_battery	79.7	83.2	61.4
MVTec AD	hazelnut	96.5	76.5	76.5	Real-IAD	plastic_nut	75.0	96.4	51.2
MVTec AD	metal_nut	98.6	86.3	61.8	Real-IAD	plastic_plug	89.4	85.6	45.4
MVTec AD	pill	98.2	84.3	80.7	Real-IAD	porcelain_doll	69.3	77.9	55.1
MVTec AD	screw	97.1	76.0	91.5	Real-IAD	regulator	73.3	72.0	50.5
MVTec AD	toothbrush	100.0	80.7	76.1	Real-IAD	switch	77.3	94.2	44.9
MVTec AD	transistor	98.7	81.5	81.8	Real-IAD	tape	87.5	94.3	56.5
MVTec AD	zipper	100.0	82.4	66.6	Real-IAD	terminalblock	80.7	71.3	51.0
VisA	candle	95.8	82.5	53.7	Real-IAD	toothbrush	87.7	98.0	56.9
VisA	capsules	91.7	98.0	58.3	Real-IAD	toy	82.0	83.7	59.0
VisA	cashew	99.7	92.6	70.2	Real-IAD	usb_adaptor	97.2	88.6	59.1
VisA	chewinggum	97.2	71.8	49.1	BTAD	01	95.1	56.7	68.3
VisA	fryum	94.3	95.6	53.5	BTAD	02	93.1	68.9	72.2
VisA	macaroni1	98.3	80.7	46.9	BTAD	03	97.7	61.3	74.3
VisA	macaroni2	96.0	85.3	47.2	MVTec AD2	can	86.8	77.7	34.0
VisA	pcb1	98.7	74.2	69.0	MVTec AD2	fabric	87.9	43.3	66.4
VisA	pcb2	97.2	75.9	50.0	MVTec AD2	fruit_jelly	91.4	56.1	36.2
VisA	pcb3	98.1	73.6	53.8	MVTec AD2	rice	83.3	84.0	17.1
VisA	pcb4	94.6	98.0	60.2	MVTec AD2	sheet_metal	78.5	61.9	63.0
VisA	pipe_fryum	100.0	93.5	65.0	MVTec AD2	vial	79.5	68.0	53.4
Real-IAD	audiojack	84.3	94.5	54.0	MVTec AD2	wallplugs	86.7	87.1	46.4
Real-IAD	bottle_cap	75.6	80.8	52.8	MVTec AD2	walnuts	85.8	76.2	34.7
MVTec AD mean		99.1	84.5	75.5	Real-IAD mean		83.1	84.9	53.9
VisA mean		96.8	85.2	56.4	BTAD mean		95.3	62.3	71.6
MVTec AD2 mean		85.0	69.3	43.9	Five-dataset mean		91.9	77.2	60.3

Table 11: Category-level MoReAD results. Dataset means match the aggregate numbers reported in the main comparison. Dataset means are computed from full-precision category metrics before rounding for display.

Dataset	Epochs	I-F1	P-AUPRO	P-AP
MVTec AD	10	98.78	79.78	68.49
VisA	10	95.72	80.97	52.93
BTAD	1	95.24	57.01	72.80

Table 12: Directly measured DINO baseline checks. MVTec AD and VisA come from the core A0 run; BTAD is a measured one-epoch multiclass baseline used to verify the additional dataset adapter.

Configuration	P-AUPRO		
	I-F1↑	@0.05↑	P-AP↑
DINO baseline	86.70	66.47	49.97
CR-MoE	87.36	68.05	51.63
CBTM	87.14	68.72	52.18
CR-MoE + CBTM	87.80	72.17	56.47

Table 13: Component transfer on Real-IAD, BTAD, and MVTec AD2.

Coefficient source	P-AUPRO			
	I-F1↑	@0.05↑	P-AP↑	APG↑
Token-wise encoder	97.45	84.85	65.70	0.094
Decoder state	96.04	77.28	55.86	0.039
Globally pooled encoder	96.72	81.14	60.35	0.066
Uniform coefficients	97.06	82.19	61.72	0.071
Frozen normal coefficients	96.93	81.76	61.08	0.069

Table 14: Controls on the CBTM coefficient channel.

Routing variant	$H(q)$	$H(p)$	KL↓	Agrec.↑	Geo↓
Hard argmin	0.18	0.24	0.214	72.6	0.136
Uniform ensemble	1.39	1.39	0.382	25.0	0.184
Single expert	0.00	0.00	0.447	44.3	0.171
Without distillation	0.43	0.28	0.231	85.8	0.115
Distilled Top- n	0.45	0.37	0.172	80.4	0.113

Table 15: Responsibility, routing, and expert-use diagnostics. Agreement is Top- n Jaccard for the last column of routing choices.

Setting	P-AUPRO		
	I-F1↑	@0.05↑	P-AP↑
$K_e = 2, \text{Top-1}, K_b = 8$	98.20	81.16	70.74
$K_e = 2, \text{Top-2}, K_b = 8$	98.11	81.22	70.82
$K_e = 4, \text{Top-1}, K_b = 8$	97.86	81.30	70.37
$K_e = 4, \text{Top-2}, K_b = 4$	98.02	81.45	70.28
$K_e = 4, \text{Top-2}, K_b = 8$	97.87	81.49	70.60
$K_e = 4, \text{Top-2}, K_b = 16$	98.30	81.42	71.08
$K_e = 4, \text{Top-4}, K_b = 8$	97.96	81.38	70.25
$K_e = 8, \text{Top-2}, K_b = 8$	98.33	81.25	70.89
$K_e = 12, \text{Top-1}, K_b = 8$	97.95	81.40	70.21
$K_e = 12, \text{Top-12}, K_b = 8$	98.02	81.34	70.41

Table 16: Numerical hyperparameter sensitivity on the VisA–Real-IAD mean.

Method	Train h	GFLOPs	Params	Memory	FPS	Latency (ms)
DINO baseline	5.8	210.62	148.01M	667.1	159.59	6.27
CR-MoE	6.6	309.00	148.41M	677.0	135.30	7.39
CBTM	6.9	310.46	149.32M	676.1	117.88	8.48
MoReAD	7.3	310.81	149.54M	699.8	102.90	9.72

Table 17: Extended training and inference efficiency. Training time is the wall-clock time for the 10-epoch MVTec multiclass run on a single RTX 5090 GPU.

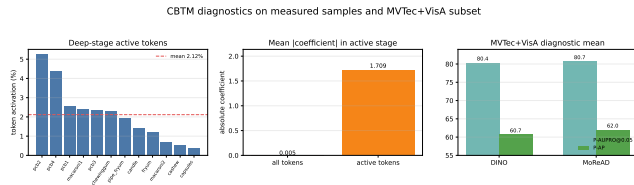
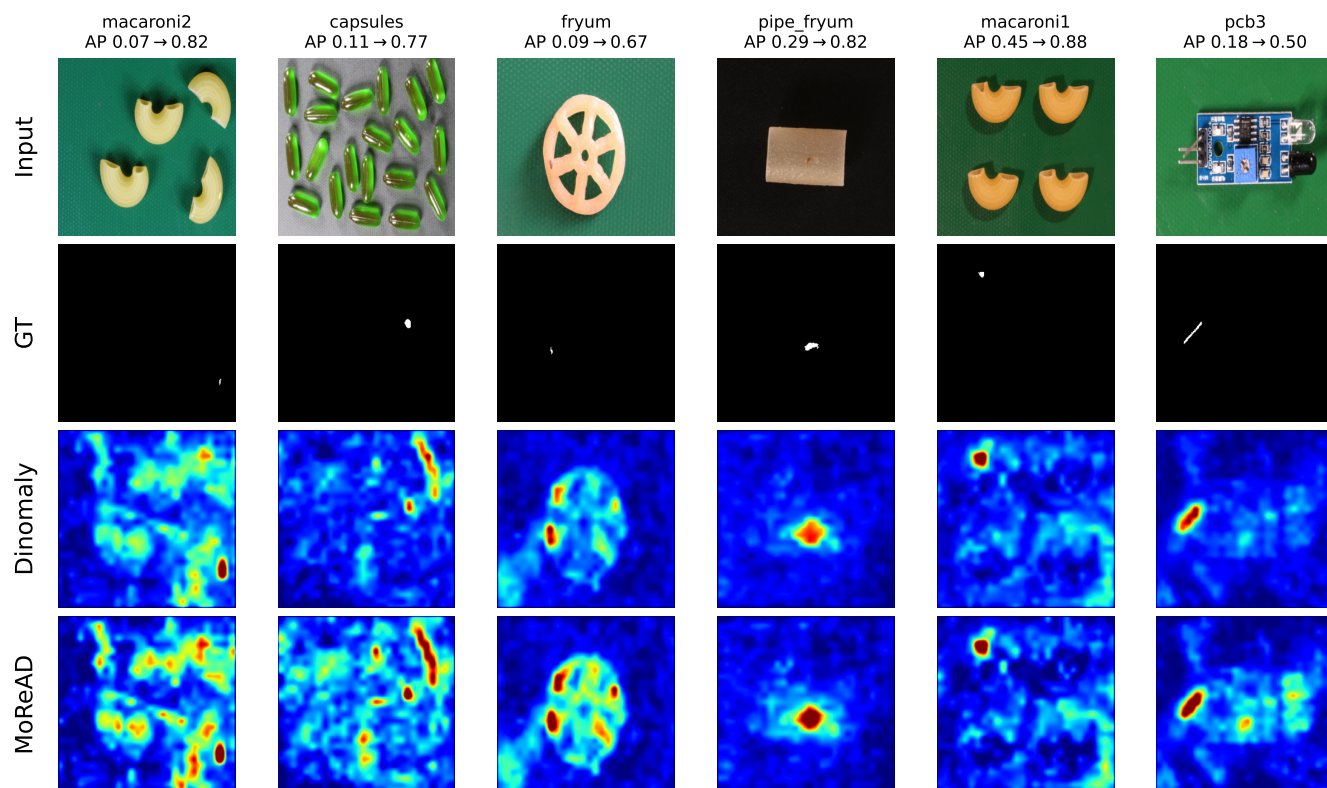


Figure 3: CBTM diagnostics from measured samples and MVTec–VisA ablations. The shallow transition is suppressed by the learned gate, while the deep transition activates sparsely with nonzero coefficients. The metric panel is a diagnostic subset summary rather than the full five-dataset result.



Real VisA improvement cases selected by largest per-sample pixel-AP gain; baseline and MoReAD maps share per-sample display scaling.

Figure 4: Representative improvement cases. Titles report the per-sample pixel-AP of Dinomaly and MoReAD, respectively.